

① 공식($a^* = \text{Cov}/\text{Var}$, $b^* = E[X] - a^*E[Y]$) 먼저 떠올리기 ② 평균·분산·공분산만 계산하면 끝 ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- LLMS 해: $\Theta^*_L = E[\Theta] + (\text{Cov}(\Theta, X)/\text{Var } X)(X - E[X]) = E[\Theta] + \rho(\sigma_\Theta/\sigma_X)(X - E[X])$
- 선형추정 $X^* = aY + b$: $a^* = \text{Cov}(X, Y)/\text{Var } Y$, $b^* = E[X] - a^*E[Y]$
- 최소 MSE: $e^*_L = (1 - \rho^2) \cdot \text{Var } X = \text{Var } X - \text{Cov}(X, Y)^2/\text{Var } Y$
- 분포 전체 불필요 — 평균·분산·공분산만 알면 설계 가능
- $\rho=0$ 이면 데이터 무용($\Theta^*_L = E[\Theta]$) / $|\rho|=1$ 이면 $e^*_L=0$ (완벽추정)
- MMSE(비선형) $x^*_M(y) = E[X|Y=y]$ vs LMSE(선형) 구분; 결합정규면 둘이 일치
- 모든 PMF/PDF는 명시 구간 밖에서 otherwise = 0

문항 목차 (총 5문항)

- [1] L12 · PSP 12.2.1 — 이산 결합PMF·E/Var/Cov·선형추정 a^*, b^*, e^*_L ·조건부 ★
- [2] L12 · PSP 12.2.2 — 신호+가우시안잡음 $R=V+X$ ·선형추정 $V^*=aR+b$ ★
- [3] L12 · PSP 12.2.3 — 이산 결합PMF· $Y^*_L = aX+b$ ·최소MSE e^*_L
- [4] L12 · PSP 12.2.4 — 연속 결합PDF·LMMSE $X^*_L(Y)$
- [5] L12 · PSP 12.2.8 — 제약형 $X^* = aY \cdot a^*$ ·최소MSE·LMSE 조건

12.2.1 ■ Random variables X and Y have joint PMF

$P_{X,Y}(x, y)$	$y = -3$	$y = -1$	$y = 1$	$y = 3$
$x = -1$	1/6	1/8	1/24	0
$x = 0$	1/12	1/12	1/12	1/12
$x = 1$	0	1/24	1/8	1/6

- Find the marginal probability mass functions $P_X(x)$ and $P_Y(y)$.
- Are X and Y independent?
- Find $E[X]$, $\text{Var}[X]$, $E[Y]$, $\text{Var}[Y]$, and $\text{Cov}[X, Y]$.
- Let $\hat{X}(Y) = aY + b$ be a linear estimator of X . Find a^* and b^* , the values of a and b that minimize the mean square error e_L .
- What is e_L^* , the minimum mean square error of the optimum linear estimate?
- Find $P_{X|Y}(x | -3)$, the conditional PMF of X given $Y = -3$.
- Find $\hat{x}_M(-3)$, the optimum (nonlinear) mean square estimator of X given $Y = -3$.
- Find the mean square error

$$e^*(-3) = E[(X - \hat{x}_M(-3))^2 | Y = -3]$$

of this estimate.

12.2.2● A telemetry voltage V , transmitted from a position sensor on a ship's rudder, is a random variable with PDF

$$f_V(v) = \begin{cases} 1/12 & -6 \leq v \leq 6, \\ 0 & \text{otherwise.} \end{cases}$$

A receiver in the ship's control room receives $R = V + X$, The random variable X is a Gaussian $(0, \sqrt{3})$ noise voltage that is independent of V . The receiver uses R to calculate a linear estimate of the telemetry voltage: $\hat{V} = aR + b$. Find

- (a) the expected received voltage $E[R]$,
- (b) the variance $\text{Var}[R]$ of the received voltage,
- (c) the covariance $\text{Cov}[V, R]$ of the transmitted and received voltages,
- (d) a^* and b^* , the optimum coefficients in the linear estimate,
- (e) e_L^* , the minimum mean square error of the estimate.

12.2.3 ● Random variables X and Y have joint PMF given by the following table:

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	$3/16$	$1/16$	0
$x = 0$	$1/6$	$1/6$	$1/6$
$x = 1$	0	$1/8$	$1/8$

We estimate Y by $\hat{Y}_L(X) = aX + b$.

- Find a and b to minimize the mean square estimation error.
- What is the minimum mean square error e_L^* ?

12.2.4■ The random variables X and Y have the joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} 2(y+x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is $\hat{X}_L(Y)$, the linear minimum mean square error estimate of X given Y ?

12.2.8 ■ For random variables X and Y , we wish to use Y to estimate X . However, our estimate must be of the form $\hat{X} = aY$.

- (a) Find a^* , the value of a that minimizes the mean square error $e = E[(X - aY)^2]$.
- (b) For $a = a^*$, what is the minimum mean square error e^* ?
- (c) Under what conditions is $\hat{X}$ the LMSE estimate of X ?